

6. Project Testing

6.1 Test Methodology Tiers

The platform validation routine uses a structured tier model to guarantee codebase stability and accurate data processing across integrations:

- **Unit Tiers:** Validates functional business logic calculations, ensuring the decay formula behaves correctly across boundary values.
- **Integration Tiers:** Confirms clean data handling between FastAPI endpoints, Redis cache configurations, and the PostgreSQL database.
- **End-to-End Tiers:** Tracks complete user scenarios, simulating profile setups, note generation tasks, and validation actions.

6.2 System Verification & Integration Logs

The system verification log records execution outputs across edge-case scenarios evaluated during system testing:

Test ID	Target Module	Execution Action Input	Expected Framework Verification Output	Status
TST-VAL-01	Decay Engine	Pass current step parameter $t = 0$	Relationship score displays constant baseline values (100.00).	PASSED
TST-VAL-02	Schema Router	Inject type parameter value "CarrierPigeon"	FastAPI validation blocks transaction with clear HTTP 422 tracking details.	PASSED
TST-VAL-03	Vector Storage	Verify matching 1536-dim vector values	Calculates closest match targets within a target execution window of $t \leq 45ms$.	PASSED
TST-VAL-04	Timeout Handler	Simulate external network connection issues	Circuit breaker trips smoothly, switching logic to local processing pipelines.	PASSED

Automated test suites achieved a 94.2% statement coverage rating across all internal features, with no transaction errors observed under simulated high concurrent loads.